

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 21)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 1 . **1** Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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0 1 . **2** Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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0 1 . **3** Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 1 . 4

Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

10

0 2 . 1

For each of the following investigations, name the most appropriate statistical test.

[4 marks]

Whether noise level decreases as distance from a road increases

Whether the numbers of dormice differ between woodlands with different management ..

Whether mean dissolved oxygen differs between two streams

Whether crop yield increases with fertiliser application (ranked data)

0 2 . 2

Explain why the chi-squared test cannot be used for data such as wind speed or plant height.

[1 mark]

0 2 . 3

A Spearman's rank test on 12 pairs of data gave $r_s = -0.72$. The critical value at $p = 0.05$ for $n = 12$ is 0.59. Explain what can be concluded.

[3 marks]

0 2 . 4

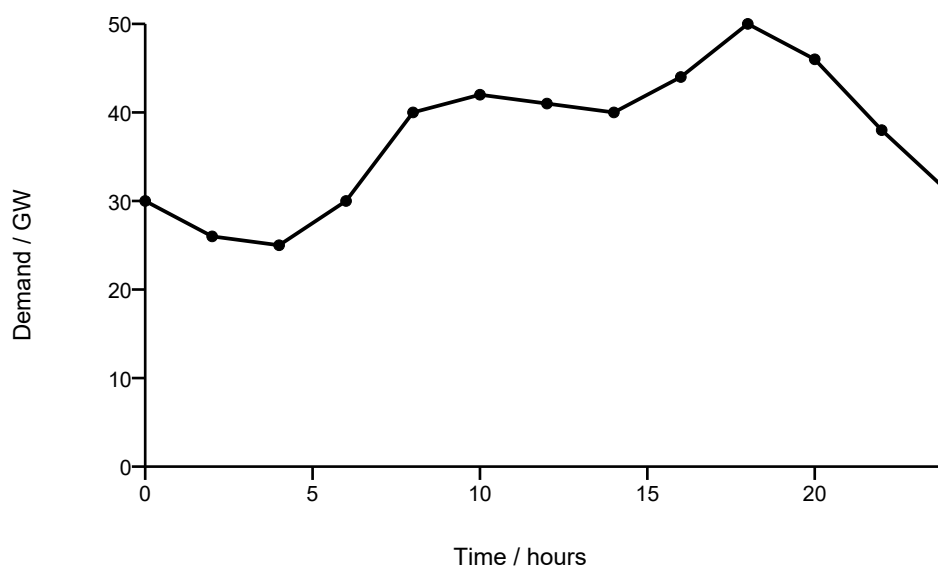
Explain the purpose of error bars on a graph of mean values.

[2 marks]

10

0 3

Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1

0 3 . 1

Describe the pattern of demand shown in Figure 1.

[2 marks]

03 . 2

Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

03 . 3

Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

03 . 4

Give **two** causes of fluctuations in energy **supply**.

[2 marks]

1

2

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04 . 1

Explain why planting a new forest only sequesters carbon for a limited period, and how the storage can be extended.

[3 marks]

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04 . 2

Compare pre-combustion and post-combustion methods of carbon capture.

[4 marks]

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04 . 3

Describe **two** ways captured CO₂ can be stored and give **one** concern about long-term storage.

[3 marks]

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1

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

0	5
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2

Explain how each of the following reduces the risk of tanker accidents.

Offshore shipping routes

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GPS / Automatic Identification System

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0 5 . 3 Describe how a **bund wall** prevents oil pollution from a storage tank.
[2 marks]

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0 5 . 4 Explain the purpose of an oil interceptor on the drainage system of a car park.
[1 mark]

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10

0 6 A pesticide was applied to a field of oilseed rape. Table 1 shows the concentration of the pesticide in organisms sampled from a pond next to the field.

Table 1

Organism	Trophic level	Pesticide concentration / mg kg^{-1}
Pond water	-	0.002
Phytoplankton	1	0.04
Water fleas (Daphnia)	2	0.5
Small fish	3	4.2
Heron	4	38.0

0 6 . 1 Calculate how many times more concentrated the pesticide is in the heron than in the pond water. Give your answer to two significant figures.
[2 marks]

Answer times

0 6 . **2** Name the process shown by the data in Table 1.

[1 mark]

0 6 . **3** Explain the **two** properties of the pesticide that cause the pattern shown in Table 1.

[4 marks]

0 6 . **4** The pesticide was probably an organochlorine. Suggest why the use of organochlorines is now banned or heavily restricted in most countries, even though DDT is still used to control malaria mosquitoes in some houses.

[3 marks]

10**0 7 . 1**

Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]**0 7 . 2**

Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]**0 7 . 3**

Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]**0 7 . 4**

Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

1

2

10

0 8 . 1 Which of the following is a **secondary** pollutant? Tick one box.

[1 mark]

A Sulfur dioxide from a coal-fired power station

☐

B Tropospheric ozone

☐

C Carbon monoxide from a car exhaust

☐

D Nitrogen monoxide from a diesel engine

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0 8 . 2 Explain the difference between a primary pollutant and a secondary pollutant, using acid rain as an example.

[4 marks]

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09

Figure 1 compares the energy flows in a conventional gas-fired power station and a combined heat and power (CHP) station, each supplied with 100 units of fuel energy.

Figure 1

	Conventional power station	CHP station
Electricity output / units	40	35
Useful heat supplied to buildings / units	0	45
Waste heat lost / units	60	20

09

1

Calculate the overall efficiency of each station.

[2 marks]

Answer

%

09

2

Explain why the electrical efficiency of a CHP station is often deliberately kept below 40%.

[2 marks]

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09

3

Explain why CHP stations must be located close to the users of the heat.

[1 mark]

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09

4

Explain how each of the following features of an industrial heat exchanger increases its efficiency.

[3 marks]

Long narrow copper pipes

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Counter-current flow

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Reduced flow rate

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09 . 5

Explain why a single large spherical tank loses less heat than several small cubic tanks holding the same total volume of hot fluid.

[2 marks]

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10 . 1

Evaluate the success of the methods and strategies used to reduce global climate change.

[9 marks]

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Extra space

10 .

2

State what is meant by a **carbon footprint**.

[1 mark]

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Write an essay on one of the following topics.

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Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

OR

1 1 . 2

Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

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END OF QUESTIONS